

# The fastest correct inverse for strong-phase LEEM: discrete Jacobi–Anger fitting, not pixel Gauss–Newton

Wilson Lau

24 August 2026

## Abstract

Linearized CTF inversion is biased for the kinematic LEEM phase object  $\psi = \exp(i\varphi \sin(kx))$  once  $\varphi = 4\pi A/\lambda_0$  is no longer small. Among four candidate inverses—Gauss–Newton on bilinear Fourier optics (FO), Jacobi–Anger discrete-mode fitting, Gerchberg–Saxton with an FO projector, and phase-diversity / MAL—only the first two are well-posed for the Yu–Lau–Altman forward model, and only the second is cheap. For 1D periodic ripples the correct unknown is a handful of Bessel coefficients inside the contrast aperture, not the  $N_{\text{ap}}$  aperture pixels. That reduces a bilinear solve of cost  $O(N_{\text{ap}}^2)$  to an  $O(M^2)$  evaluation with  $M = 2\lfloor q_{\text{ap}}\Lambda \rfloor + 1 \ll N_{\text{ap}}$ . For general 2D topography the same reduction is unavailable; the fastest still-correct method is then damped Gauss–Newton on bilinear FO, initialized by the linearized CTF. Gerchberg–Saxton is not a distinct correct algorithm under partial coherence, and MAL / phase diversity is the same Gauss–Newton residual stacked over extra defoci (needed only when aberrations are unknown or CTF zeros must be filled). We recommend a two-stage pipeline and a precise skip test for stage 2, and we identify the Lean theorems that make this inverse formalizable.

## 1 The inverse that the forward model actually poses

Image intensity in the Yu–Lau–Altman FO model is the bilinear form

$$I(\mathbf{r}) = \iint \Psi(\mathbf{q}) \Psi^*(\mathbf{q}') R_{\text{FO}}(\mathbf{q}, \mathbf{q}', \Delta z) e^{2\pi i(\mathbf{q}-\mathbf{q}')\cdot\mathbf{r}} d^2q d^2q', \quad (1)$$

with  $R_{\text{FO}}$  supported on the contrast-aperture disk  $|\mathbf{q}|, |\mathbf{q}'| \leq q_{\text{ap}}$  [1]. A separable CTF kernel  $R_{\text{CTF}}$  agrees with  $R_{\text{FO}}$  on the axes  $\mathbf{q} = \mathbf{0}$  or  $\mathbf{q}' = \mathbf{0}$  (already a Lean theorem: `R_FO_eq_R_CTF_axis`), and  $R_{\text{CTF}} = R_{\text{FO}} \Gamma_C \Gamma_S$  off axis. Linearized (weak-phase) inversion uses only those axes: if  $\psi = \sigma(1 + i\phi) + O(\phi^2)$ , the contrast at frequency  $\mathbf{q}$  is proportional to  $R_{\text{FO}}(\mathbf{q}, \mathbf{0})$ , where  $\Gamma_C \Gamma_S = 1$ . The linear inverse is therefore the correct first-order Fréchet derivative of (1) even under partial coherence.

It is biased as soon as quadratic products  $\Psi(\mathbf{q})\Psi^*(\mathbf{q}')$  with *both*  $\mathbf{q}, \mathbf{q}' \neq \mathbf{0}$  rise above noise. For a sinusoidal corrugation those products are not a perturbation: they are the image.

## 2 Object classes, $N_{\text{ap}}$ and $M$

**Definition 2.1** (Aperture degrees of freedom). Let  $L$  be the field of view and  $\Delta q = 1/L$ . Then

$$N_{\text{ap}} := \#\{\text{sampled } \mathbf{q} : |\mathbf{q}| \leq q_{\text{ap}}\} = \begin{cases} 2\lfloor q_{\text{ap}}L \rfloor + 1, & \text{1D lineout,} \\ \pi(q_{\text{ap}}L)^2, & \text{2D circular aperture.} \end{cases} \quad (2)$$

On a Nyquist grid  $N = 2q_{\text{ap}}L$  one has  $N_{\text{ap}} = \pi N^2/4$  in 2D and  $N_{\text{ap}} = N$  in 1D.

IBM-style parameters [3, 1] ( $E_0 = 10$  eV,  $\lambda_0 \simeq 0.388$  nm, column  $\lambda \simeq 0.010$  nm, information limit  $\sim 1$ – $2$  nm) give  $q_{\text{ap}} \sim 0.5$ – $1$  nm $^{-1}$ . A  $2\text{ }\mu\text{m}$  field of view then has  $N_{\text{ap}}^{\text{1D}} \sim 2 \times 10^3$  and  $N_{\text{ap}}^{\text{2D}} \sim 3 \times 10^6$  (or  $N_{\text{ap}}^{\text{2D}} \sim 5 \times 10^4$  on a  $256^2$  Nyquist patch). Naive assembly of (1) is  $O(N_{\text{ap}}^2)$ : this is why FO is slow compared with a CTF FFT of cost  $O(N \log N)$  [2, 1].

**Definition 2.2** (Bessel modes of a 1D ripple). For  $\psi(x) = \exp(i\varphi \sin(2\pi x/\Lambda))$  the Jacobi–Anger theorem (Lean: `jacobi_anger`) gives

$$\psi(x) = \sum_{n \in \mathbb{Z}} J_n(\varphi) e^{2\pi i n x / \Lambda}, \quad q_n = n / \Lambda. \quad (3)$$

The aperture keeps only

$$M := 2n_{\text{ap}} + 1, \quad n_{\text{ap}} = \lfloor \min(q_{\text{ap}}\Lambda, n_*(\varphi)) \rfloor, \quad (4)$$

where  $n_*(\varphi) \simeq |\varphi| + 2|\varphi|^{1/3}$  is the Olver turning point of  $J_n(\varphi)$  (practically  $|\varphi| + 3$ ). Bessel decay, not the pixel grid, sets  $M$  once  $q_{\text{ap}}\Lambda > |\varphi|$ .

Kinematic amplitude  $\varphi = 4\pi A/\lambda_0$  is large for ordinary LEEM topography:  $A = 0.2$  nm already gives  $\varphi \simeq 6.5$ ; a 1 nm wrinkle gives  $\varphi \simeq 32$ . Typical counts:

| Object                          | $\Lambda$  | $\varphi$     | $M$             | $N_{\text{ap}}$ (1D, $L = 2\text{ }\mu\text{m}$ ) |
|---------------------------------|------------|---------------|-----------------|---------------------------------------------------|
| thermal graphene ripple         | 10 nm      | $\sim 2$      | $\sim 11$       | $\sim 2 \times 10^3$                              |
| LEEM wrinkle / Yu test sinusoid | 100–500 nm | $10$ – $10^2$ | $10^2$ – $10^3$ | $\sim 2 \times 10^3$                              |
| general 1D topography           | —          | —             | $N_{\text{ap}}$ | $\sim 2 \times 10^3$                              |
| general 2D topography, $256^2$  | —          | —             | $N_{\text{ap}}$ | $N_{\text{ap}} \sim 5 \times 10^4$                |

The Yu sinusoidal object therefore lives in a subspace of dimension  $M \sim 10$ – $10^3$ , not  $N_{\text{ap}}$  and not the pixel count  $N$ .

### 3 Four candidates

#### A. Nonlinear Gauss–Newton on bilinear FO

Unknown: the aperture spectrum  $\Psi$  (or a real phase  $\phi(\mathbf{r})$  of length  $N_{\text{ap}}$ ). Residual  $I_{\text{FO}}(\Psi) - I_{\text{obs}}$ . Each iteration evaluates (1) in  $O(N_{\text{ap}}^2)$  (fill the difference-frequency histogram, then one FFT) and applies the adjoint of the bilinear map. A damped Gauss–Newton / Levenberg–Marquardt step with inner CG costs

$$O(k n_{\text{CG}} N_{\text{ap}}^2) \quad (k \sim 5\text{--}20 \text{ outer iterations if the linear init is in basin}). \quad (5)$$

Initialized by linearized CTF this is *locally* the correct maximum-likelihood estimator for Gaussian noise and known optics. It is not globally unique from a single defocus (CTF zeros; bilinear phase retrieval). For a sinusoid it is correct but spends  $N_{\text{ap}}^2/M^2$  useless work on empty Fourier bins.

#### B. Jacobi–Anger discrete-mode fitting

Substitute (3) into (1) and drop modes with  $|n| > q_{\text{ap}}\Lambda$ :

$$I(x) = \sum_{n,m=-n_{\text{ap}}}^{n_{\text{ap}}} J_n(\varphi) J_m(\varphi) R_{\text{FO}}(q_n, q_m, \Delta z) e^{2\pi i (q_n - q_m)x}. \quad (6)$$

If the object is known to be a single sinusoid, the unknown is  $(\varphi, x_0, \sigma)$  — three real numbers — or even  $\varphi$  alone once the period  $\Lambda$  is read from the fundamental peak. Each residual costs  $O(M^2)$ ,

independently of  $N$  after the  $2n_{\text{ap}}$  measured harmonics of  $I$  have been extracted ( $O(N \log N)$  once).

Unconstrained  $\{c_n\}$  with  $c_n = J_n(\varphi)$  *not* imposed is a bilinear phase-retrieval problem of size  $M$  and is the wrong encoding: the physics is a one-parameter Bessel curve, not  $M$  free complex amplitudes. Least squares *on*  $J_n(\varphi)$  means the parametric map  $\varphi \mapsto (J_n(\varphi))_n$ .

Sign of  $\varphi$ :  $J_n(-\varphi) = (-1)^n J_n(\varphi)$ , so odd intensity harmonics flip with  $\text{sign}(\varphi)$ . Convex versus concave ripples are identifiable. Local uniqueness fails only at isolated Bessel extrema, which a 1D line search on  $\varphi$  skips.

This is exact for the object class that makes linearized CTF fail in Yu *et al.* [2, 1], and it is already half-formalized (`jacobi_anger`, `besselJ_bound`).

### C. Gerchberg–Saxton with an FO projector

Classical GS replaces moduli in two planes linked by a unitary FFT. Under partial coherence,  $I$  is *not*  $|\psi_{\text{img}}|^2$  of a single coherent wave: the correct intensity constraint is the quadratic set  $\{\psi : (1) = I_{\text{obs}}\}$ . The metric projector onto that set *is* a Gauss–Newton step on the FO residual. So a correctly specified “GS+FO projector” is A with worse globalization (stagnation, no damping, no Gauss–Newton Hessian). A naïve modulus-replacement projector is the coherent CTF model and is biased precisely when  $\Gamma_C \Gamma_S \neq 1$  off axis. GS is therefore either incorrect or a slower, less stable rewrite of A. It is a poor Lean target: no uniqueness theorem, no closed residual.

### D. Phase diversity / MAL

Phase diversity [4] jointly estimates object and aberrations from  $N_f$  known-diversity images. MAL [5] (Coene–Thust maximum-likelihood, typically initialized by the paraboloid method PAM) is Gauss–Newton on the *stacked* FO residual of a focal series. Cost

$$O(N_f k n_{\text{CG}} N_{\text{ap}}^2). \quad (7)$$

Statistically efficient, fills CTF zeros, and is the right estimator when  $\chi$  is unknown. When optics are known and the object is a 1D sinusoid, extra defoci are unused degrees of freedom: one non-zero CTF sample at  $q = 1/\Lambda$  already identifies  $\varphi$  in (6). D is A with more data, not a faster inverse.

## 4 Verdict: fastest still-correct

| Class                                      | Fastest correct                                                                                                 | Why the others lose                                                                                                             |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 1D periodic ripple $e^{i\varphi \sin(kx)}$ | <b>B</b> (Jacobi–Anger LS)                                                                                      | A is the same residual on a wasteful pixel basis; C is A or is misspecified; D multiplies the cost by $N_f$ for no new unknown. |
| 1D general $\phi(x)$                       | A on $N_{\text{ap}}^{\text{1D}}$ samples, or B’ (Fourier series of $\phi$ with $P \ll N_{\text{ap}}$ harmonics) | still $M \sim 2Pn_* \ll N_{\text{ap}}$ if $\phi$ is smooth.                                                                     |
| 2D general $\phi(x, y)$                    | <b>A</b> , linear CTF init; D only if aberrations unknown or a single defocus sits on a CTF zero                | B does not apply (no discrete Bessel ladder); C as above.                                                                       |

*Pick for the Lean/PDF formalization of Yu’s sinusoid:* **B**, with linearized CTF as the cheap initializer (stage 1) and with A recorded as the 2D generalisation.

## 5 Two-stage method

**Stage 1 (always).** Linear CTF / Wiener inversion on the  $q' = 0$  slice, cost  $O(N \log N)$  or  $O(n_{\text{ap}})$  after extracting harmonics. For a sinusoid: read  $\Lambda$  from the fundamental of  $I$ , divide the complex amplitude at  $q = 1/\Lambda$  by  $R_{\text{FO}}(q, 0, \Delta z)$  (equals  $R_{\text{CTF}}(q, 0, \Delta z)$ ), obtain an apparent amplitude  $\varphi_{\text{lin}}$ . This  $\varphi_{\text{lin}}$  tracks  $2J_1(\varphi)$  rather than  $\varphi$  (standard weak-phase bias). Compute the FO residual  $r_1 = \|I_{\text{FO}}(\varphi_{\text{lin}}) - I_{\text{obs}}\|$ .

**Stage 2 (1D periodic).** Scalar (or 3-parameter) Gauss–Newton / golden-section search on (6):

$$\min_{\varphi, x_0} \sum_x \left| I_{\text{obs}}(x) - I_{\text{FO}}(\{J_n(\varphi)\}) \right|^2. \quad (8)$$

Jacobian in  $\varphi$  is analytic:  $J'_n(\varphi) = \frac{1}{2}(J_{n-1}(\varphi) - J_{n+1}(\varphi))$ . Cost  $O(kM^2)$  with  $k \sim 5\text{--}20$ . This is *A restricted to the Jacobi–Anger manifold*, which is why it stays correct and becomes cheap.

**Stage 2 (2D general).** A few damped Gauss–Newton iterations on (1) in the pixel (or aperture) basis, initialized by stage 1. Optional: CTF Jacobian with FO residual (inexact GN). Use D / a focal series only if stage 1 hits a CTF zero or  $\chi$  is part of the unknown.

**When stage 2 is unnecessary.** The linear term of (1) lives on the axes, where  $R_{\text{FO}} = R_{\text{CTF}}$ . Quadratic (and higher) Bessel products are the whole bias. Skip stage 2 when they sit below noise:

1. **Weak phase.**  $\max|\varphi| \lesssim 0.3$  (i.e.  $A \lesssim 0.01$  nm at  $E_0 = 10$  eV). Then  $J_1(\varphi) = \varphi/2 + O(\varphi^3)$  and  $|J_2/J_1| \simeq |\varphi|/4 \lesssim 0.08$ .
2. **Aperture already kills higher orders, and  $\varphi$  is still moderate.** If  $q_{\text{ap}}\Lambda < 2$  then  $M = 3$  ( $n = 0, \pm 1$ ). The remaining nonlinearity is  $J_0(\varphi) \approx 1 - \varphi^2/4$  and  $J_1(\varphi) \approx \varphi/2$ . Skip only if that  $\varphi^2$  error is below noise, not merely because  $M = 3$ .
3.  **$\chi^2$  test on the FO residual of stage 1.** If  $r_1^2/\sigma^2$  is consistent with the noise model, the quadratic terms are undetectable; stage 2 cannot improve the estimate. This is the operational skip, and it covers “ $\Gamma_C\Gamma_S \approx 1$  on populated pairs” automatically: those pairs only enter at quadratic order.

Do *not* skip stage 2 merely because  $\Gamma_C\Gamma_S \approx 1$ . On the axes  $\Gamma = 1$  always; the linear inverse never saw the off-axis envelopes. Strong-phase bias is a property of  $J_n(\varphi)$ , not of partial coherence. Partial coherence changes the *weights*  $R_{\text{FO}}(q_n, q_m)$  in (6) and so must sit in the *forward* model of stage 2, but it does not by itself decide whether stage 2 is needed.

## 6 What to encode in Lean (and what not)

Formalize the *forward identity* of B and the linearization bias; do not formalize GS iterates or a MAL driver.

1. Discrete spectrum of  $e^{i\varphi \sin(kx)}$  (corollary of `jacobi_anger`): support on  $q_n = n/\Lambda$ .
2. Aperture section:  $M = 2\lfloor q_{\text{ap}}\Lambda \rfloor + 1$  modes; Bessel tail  $\sum_{|n|>N} |J_n(\varphi)| \leq 2(|\varphi| + \varphi^2)/N$  from `besselJ_bound`.
3. Finite bilinear image (6) as a double sum of length  $M^2$  (complexity theorem: `card_modePairs`).

4. Linearization: the Fréchet derivative of (1) at  $\varphi = 0$  is the axis operator  $R_{\text{CTF}} = R_{\text{FO}}(\cdot, 0)$  (reuse `R_FO_eq_R_CTF_axis`).
5. Stage-2 skip: if  $\sum_{|n| \geq 2} |J_n(\varphi)|^2 < \varepsilon$  then the linear inverse's relative error is  $O(\varepsilon + \varphi^2)$ .

Items C and D are algorithms, not identities. A in 2D is local convergence of Gauss–Newton on a bilinear map — true, but it does not use any LEEM-specific theorem beyond (1). The LEEM-specific, already half-written content is B.

## 7 Conclusion

For the object that the FO/CTF papers actually use to exhibit linearization failure — 1D sinusoidal / strong-phase ripples — the fastest still-correct inverse is Jacobi–Anger discrete-mode fitting (B), not pixel Gauss–Newton, not GS, not MAL. Cost  $O(M^2)$  with  $M = 2\lfloor \min(q_{\text{ap}}\Lambda, |\varphi| + 3) \rfloor + 1$ , versus  $O(N_{\text{ap}}^2)$  for A and  $O(N_{\text{f}}N_{\text{ap}}^2)$  for D. For 2D general objects B does not apply; use linearized CTF initialization plus a few Gauss–Newton steps on bilinear FO (A), and invoke phase diversity / MAL only when the optics are unknown or a single defocus is CTF-blind. Stage 2 of the two-stage pipeline is unnecessary when the FO residual of the linear reconstruction is at the noise floor, equivalently when  $\max|\varphi| \lesssim 0.3$  for typical LEEM SNR.

## References

- [1] K. M. Yu, K. L. W. Lau, M. S. Altman, *Fourier optics of image formation in aberration-corrected LEEM*, Ultramicroscopy **200** (2019) 160–168.
- [2] K. M. Yu, A. Locatelli, M. S. Altman, *Comparing Fourier optics and contrast transfer function modeling of image formation in low energy electron microscopy*, Ultramicroscopy **183** (2017) 109–116.
- [3] S. M. Schramm, A. B. Pang, M. S. Altman, R. M. Tromp, Ultramicroscopy **115** (2012) 88–108.
- [4] R. G. Paxman, T. J. Schulz, J. R. Fienup, J. Opt. Soc. Am. A **9** (1992) 1072–1085.
- [5] W. M. J. Coene, A. Thust, M. Op de Beeck, D. Van Dyck, Ultramicroscopy **64** (1996) 109–135.